

Area of Rectangle, Triangle, Parallelogram & Trapezoid

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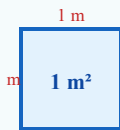
Standard: CCSS.MATH.CONTENT.6.G.A.1 — Find the area of right triangles, special quadrilaterals, and polygons by composing into rectangles or decomposing into triangles.

★ Concept #1 — What does "m²" really mean?

1 m² = the area enclosed by a 1 m × 1 m square. Look at the multiplication closely:

$$\underbrace{1}_{\text{hidden}} \times \underbrace{1}_{\text{hidden}} = 1 \quad \text{and} \quad m \times m = m^2$$

So when you write **5 m × 6 m = 30 m²**, the unit isn't decoration — it *comes from the multiplication itself*. Numbers multiply with numbers, units multiply with units.



A square that is **1 m wide** and **1 m tall** holds exactly **1 m²** of area. Bigger shapes? Just count how many of those 1 m² squares fit inside.

★ Concept #2 — Multiplication IS area (Rectangle)

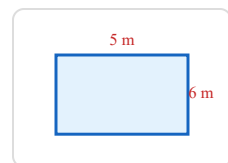
A rectangle of base b and height h holds $b \times h$ unit squares. That product — number AND unit — is the

area. $\text{Area}_{\text{rect}} = b \times h$

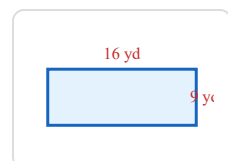
Part A — Rectangles & Squares

Find the area. Write the answer **with units**. Remember: $m \times m = m^2$, $\text{in} \times \text{in} = \text{in}^2$, ...

- 1) A rectangle has base **5 m** and height **6 m**. Find the area. Area =



- 2) A rectangle has length **16 yd** and width **9 yd**. Find the area. Area =



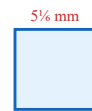
- 3) A rectangle is **3.5 cm** by **8 cm**. Find the area. Area = _____



- 4) A **square** has side **12 in**. Find the area. ($s \times s = s^2$) Area = _____

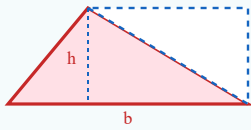


- 5) A square has side $5\frac{1}{6}$ **mm**. Find the area as a mixed number. Area = _____



★ Concept #3 — Triangle is HALF of a rectangle/parallelogram

Take any triangle. Make a copy and flip it upside-down → the two pieces form a parallelogram of the same base & height. So **one triangle is half of that**.



The dashed triangle is the "flipped copy". Together they make a rectangle of

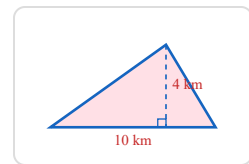
$b \times h$. So: $\text{Area}_{\text{tri}} = (b \times h) / 2$ **Important:** h is the *perpendicular* height

— NOT the slanted side.

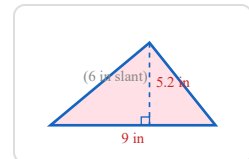
Part B — Triangles

Use the perpendicular height (drawn with a dashed line or with a right-angle mark).

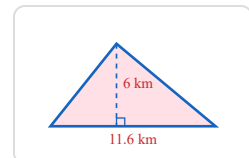
- 6) Triangle with base **10 km** and perpendicular height **4 km**. Area = _____



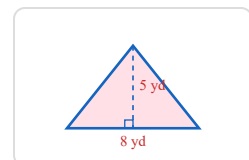
- 7) Triangle with base **9 in**, perpendicular height **5.2 in** (the slanted side is 6 in — ignore it). Area = _____



- 8) Triangle with base **11.6 km** and height **6 km**. Area = _____

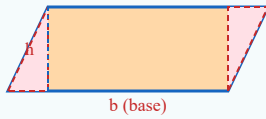


- 9) Triangle with base **8 yd** and height **5 yd**. Area = _____



★ Concept #4 — Parallelogram = Rectangle (in disguise)

Slice the slanted triangle off one end of a parallelogram, slide it to the other end → it becomes a **rectangle** with the same base and the same perpendicular height. So:

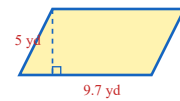


$\text{Area}_{\text{par}} = b \times h$ Same as a rectangle! Just be sure h is the perpendicular height — not the slanted side.

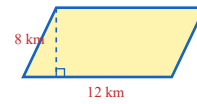
Part C — Parallelograms

Use the perpendicular height. The slanted side is a distractor — ignore it.

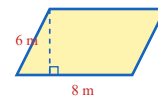
- 10) Parallelogram with base **9.7 yd**, height **5 yd** (slant side 5.8 yd given but not needed). Area = _____



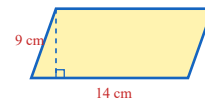
- 11) Parallelogram with base **12 km**, height **8 km**. Area = _____



- 12) Parallelogram with base **8 m**, height **6 m**. Area = _____

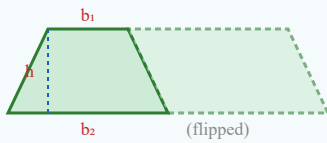


- 13) Parallelogram with base **14 cm**, height **9 cm**. Area = _____



★ **Concept #5 — Trapezoid: DON'T memorize, DRAW it**

Take any trapezoid. Make an identical copy, flip it upside down, and place it next to the original → the two together form a **parallelogram** with base $(b_1 + b_2)$ and height h . One trapezoid is half of that.



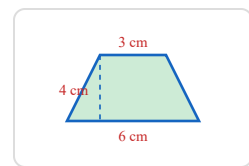
The parallelogram has base $b_1 + b_2$ and height h , so its area is $(b_1 + b_2) \cdot h$. One trapezoid is half:

$$\text{Area}_{\text{trap}} = (b_1 + b_2) \times h / 2$$

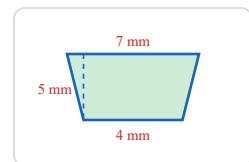
Part D — Trapezoids

Identify the two parallel sides (b_1 and b_2) and the perpendicular height h .

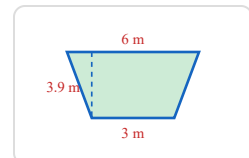
- 14) Trapezoid with $b_1 = 3 \text{ cm}$, $b_2 = 6 \text{ cm}$, height $h = 4 \text{ cm}$. Area =



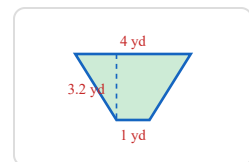
- 15) Trapezoid with $b_1 = 7 \text{ mm}$, $b_2 = 4 \text{ mm}$, height $h = 5 \text{ mm}$. Area =



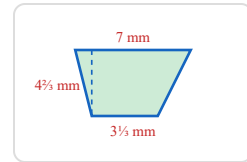
- 16) Trapezoid with $b_1 = 6 \text{ m}$, $b_2 = 3 \text{ m}$, height $h = 3.9 \text{ m}$. Area =



- 17) Trapezoid with $b_1 = 4 \text{ yd}$, $b_2 = 1 \text{ yd}$, height $h = 3.2 \text{ yd}$. Area =



18) Trapezoid with $b_1 = 7 \text{ mm}$, $b_2 = 3\frac{1}{3} \text{ mm}$, height $h = 4\frac{2}{3} \text{ mm}$. Give the answer as a mixed number. Area = _____



Part E — Word Problems

Read carefully. Draw the picture. Concept first, equation second.

- 19) A trapezoidal garden has area **96 cm²** and height **6 cm**. The longer base is **4 cm longer** than the shorter base. Find the lengths of both bases (b_1 and b_2).

$b_1 =$ _____ $b_2 =$ _____

- 20) Kenny's classroom floor is a parallelogram with base **14 m** and perpendicular height **5 m**. How many square meters of carpet does Kenny need to cover the whole floor? Area = _____
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